# Information System Design and Software Engineering

## Introduction

Ashek Seum

ashek.seum@seu.edu.bd

# Course Objectives

- To help students to **develop skills** that will enable them to construct **software of high quality**.
- Software that is **reliable**, and that is reasonably easy to understand, modify and maintain.

# Recommended Book(s)

**Software Engineering (9th Edition)**
Author: **Ian Sommerville**
Published by Addison Wesley, 2010

**Software Engineering: A Practitioner's Approach (7th Edition)**
Author: **Roger S. Pressman**
Published by McGraw-Hill College

**Systems Analysis and Design (8th Edition)**
Author: **Kenneth Kendall**
Published by Pearson Education Limited

# What is Software?

We can define it as:

- ***Instructions*** (Computer Programs) that when executed provide
  - Desired features
  - Functions and
  - Performance
- ***The data*** on which the program operates is also considered as a part of the software.
- ***Documents*** that describe the use of the Programs.

# Role of Software

- A person might be involved with software more than 100 times, often  without even realizing that.
- We might use a computer :-
  - To check email or weather
  - To play games
  - To watch movies
- Traffic signals are now computer controlled
- Our bank system is highly computerized
- Hotel arrangements will be handled by computers and software
- Air plane & Air traffic control is highly computerized
- Medical Instruments and diagnostic procedures are now computerized

# Attributes of Good Software

- Should deliver the required functionality and performance to the user.
- Reliability
- Good user-interface
- Should be maintainable and reusable

# Categories of Computer Software

- System Software
- Application Software
- Engineering/Scientific Software
- Embedded Software
- Web Applications
- Computer Programming tools
- Artificial Intelligence Software

# Software Products

- **Generic products**
  - Stand-alone systems that are marketed and sold to **any customer** who wishes to buy them.
  - **Examples** – PC software such as editing, graphics programs, project management tools, databases and drawing packages.





Microsoft Office



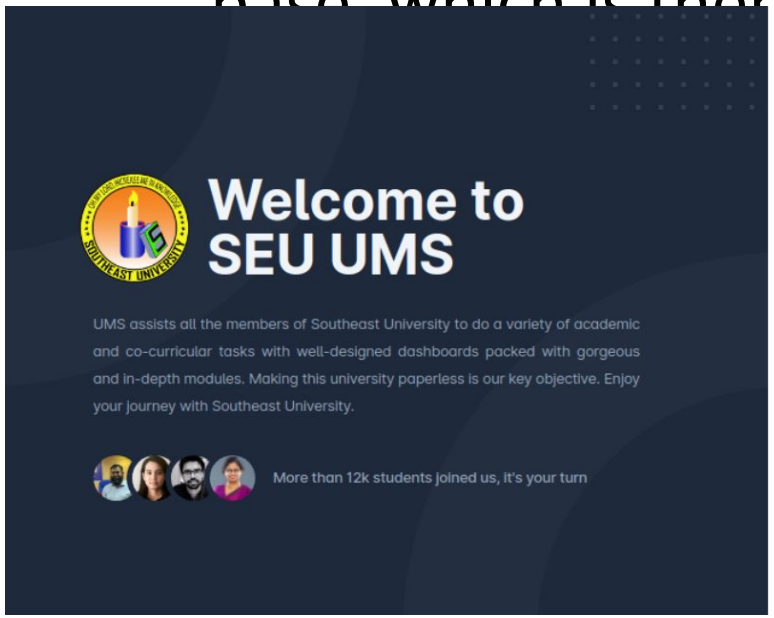
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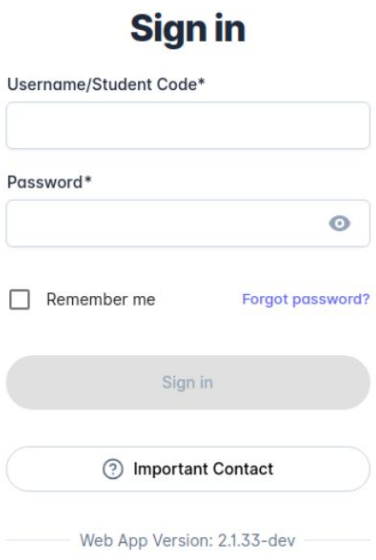


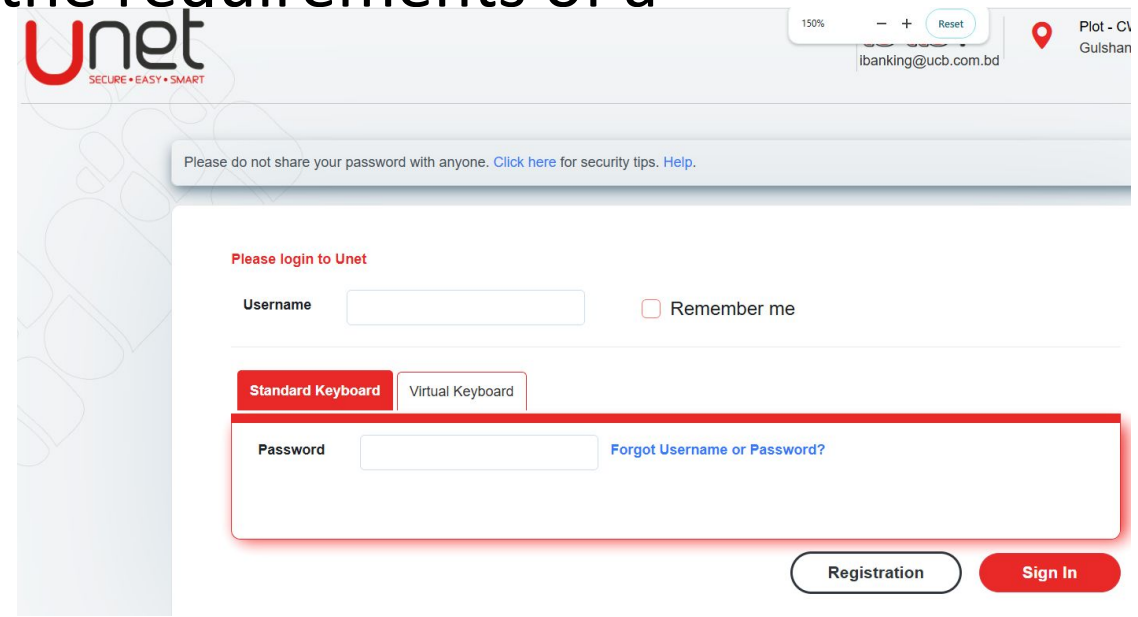
Autodesk AutoCAD

# Software Products

- **Customized products**
  - Software that is commissioned by **a specific customer** to meet their own needs.
  - **Examples**–systems written to support a particular business process.
- More and more systems are being built with a generic product as a base, which is then adapted to suit the requirements of a



# Software Crises

- In the 1960s, software suffered from a problem, which we call the ***Software Crisis***.
- The crisis manifested itself in several ways:
  - Projects running **over-budget**
  - Projects running **over-time**
  - Software was **very inefficient**
  - Software was of **low quality**
  - Software often **did not meet requirements**
  - Projects were unmanageable and **code difficult to maintain**
  - Software was **never delivered**

# Software Crises (contd..)

- **A conference was held in 1968 in which the term “software crisis” was introduced**
- Software engineering is the result of the software crisis when people realized that it was **not possible to construct complex software** using the techniques applicable in 1960s.
- An important result of this was that people realized that **just coding was not enough**; we also need to apply engineering principles.

# Software Crises (contd..)

“As long as there were no machines, programming was no problem at all; when we had a few weak computers, programming became a mild problem, and now we have gigantic computers, programming has become an equally gigantic problem.”

- Edsger Dijkstra, 1972

# What is Software Engineering?

- All aspects of software production "**Software Engineering**" is not just concerned with the **technical processes** of software development but also with activities such as **software project management** and with the development of tools, methods and theories to support software production.

- Computer Science vs. Software Engineering
  - Computer science focuses on theory and fundamentals; software engineering is concerned with the practicalities of developing and delivering useful software.
- System engineering vs. Software Engineering
  - System engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering. Software engineering is part of this.

# Software Costs

- Software costs often dominate computer system costs. The costs of software on a PC are often greater than the hardware cost.
- Software costs **more to maintain** than it does to develop. For systems with a long life, maintenance costs may be several times development costs.
- Software engineering is concerned with cost-effective software development.
- Roughly **60%** of costs are development costs, **40%** are testing costs.
- Costs vary depending on the type of system being developed and the requirements of system

# Software Engineering Ethics

- "Software engineers are those who contribute by direct participation or by teaching, to the analysis, specification, design, development,
  certification, maintenance, and testing of software systems."
  - Confidentiality
  - Competence
  - Intellectual property rights
  - Computer misuse
- Software engineers shall act consistently with the public interest.

# END

